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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

EE2004-1 – MODERN SWITCHGER AND PROTECTION

(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which material is commonly used as a fusing element in high rupturing capacity (HRC) fuses?
A) Copper
B) Aluminum
C) Ceramic
D) Silver
2. The primary function of a load breaking switch in a power system is to:
A) Provide back-up protection.
B) Measure power consumption.
C) Isolate faulty equipment located far away from circuit breaker.
D) Regulate voltage.
3. The life span of fault current is equal to
A) Pre-arcing time of the fuse.
B) Arcing time of the fuse.
C) Sum of arcing and pre-arcing times of the fuse.
D) Difference between pre-arcing time and arcing time of the fuse.
4. Which among the following statements is true?
A) Pre-arcing time of back-up fuse is longer than operating time of primary fuse.
B) Pre-arcing time of primary fuse is longer than operating time of back-up fuse.
C) Pre-arcing time of primary fuse is longer than pre-arcing time of back-up fuse.
D) Operating time of primary fuse is longer than operating time of back-up fuse.
5. In a non-directional overcurrent relay, if the direction of current is reversed, operating torque
A) Reverses direction.
B) Becomes zero.
C) Does not change direction.
D) May or may not change direction.
6. In a practical directional overcurrent relay, a directional unit and a non-directional unit are used in conjunction as
A) Non-directional unit has low torque at the time of fault.
B) These together have less weight.
C) Directional unit has low torque at the time of fault due to dip in voltage.
D) These have less restraining torque only when used together.
7. In an electromechanical overcurrent relay, as the set current is increased
A) Number of coil turns included in the circuit decreases.
B) Restraining torque increases.
C) Disc starts rotating faster.
D) Electromagnet saturates.
8. In the directional characteristics of a directional overcurrent relay,
A) The operating region has an area half that of the restraining region.
B) The operating region has an area double that of the restraining region.
C) The operating region has an area equal to that of the restraining region.
D) The operating region has an area four times that of the restraining region.
9. A hold capacitor in a digital relay:
A) Functions as a differentiator.
B) Functions as an integrator.
C) Retains the sample until analog-to-digital conversion is completed.
D) Serves as a filter.

10. Which of the following statements is true?
 A) A biased differential relay mal-operates during a heavy external fault whereas a simple differential relay does not.
 B) Both simple and biased differential relays mal-operate during a heavy external fault.
 C) Neither simple differential relay nor biased differential relay mal-operates during a heavy external fault.
 D) A simple differential relay mal-operates during a heavy external fault whereas a biased differential relay does not.
11. The following relay can detect a fault in its incipient stage:
 A) Overcurrent relay
 B) Reactance relay
 C) Impedance relay
 D) Buchholz relay
12. A restricted earth fault relay used for alternator protection
 A) Provides protection against earth fault inside and outside the alternator stator.
 B) Provides protection against earth fault and phase-phase fault inside alternator stator.
 C) Provides protection only against earth fault inside the alternator stator.
 D) Provides protection only against phase-phase fault inside alternator stator.
13. The protection method commonly used against bus faults in power systems is
 A) Overcurrent protection
 B) Distance protection
 C) Undercurrent protection
 D) Differential protection
14. Carrier current protection is normally used for the protection of
 A) Induction motors.
 B) Alternators.
 C) Transmission lines.
 D) Transformers.
15. In biased differential protection of a star/delta transformer,
 A) CT's on the star side are connected in delta and CT's on the delta side are connected in star.
 B) CT's on both star and delta side are connected in delta
 C) CT's on both star and delta side are connected in star.
 D) CT's on the star side are connected in star and CT's on the delta side are connected in delta.
16. The reach of the following distance relay is not affected by fault resistance:
 A) Impedance relay
 B) Mho relay
 C) Offset mho relay
 D) Reactance relay
17. What is common cause of overvoltage in electrical systems?
 A) Transformer tap-changer operation
 B) Transformer testing
 C) Load shedding
 D) Lightning and switching
18. What initiates the arc during circuit breaking?
 A) Field emission
 B) Inflammable material
 C) Thermionic emission
 D) Fire lighted from outside
19. What makes DC circuit breaking more challenging than AC circuit breaking?
 A) No zero crossings to interrupt current
 B) Higher energy losses
 C) Lower voltage levels
 D) Circuit resistance
20. Which type of system earthing limits earth fault current without causing active power loss?
 A) Solid earthing
 B) Resistance earthing
 C) Reactance earthing
 D) Both solid and resistance earthing

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive the fuse law.	4	L*2	1	1,2
	b) Define the following for an electrical fuse: (i) rated current (ii) minimum fusing current (iii) fusing factor (iv) operating time.	4	L2	1	1,2
	c) Describe the construction and working of a directional overcurrent relay with a neat circuit diagram. Why does it require two units?	8	L4	2	1,2
2.	a) Elaborate the construction and working of HRC fuse with a neat figure.	6	L2	1	1,2
	b) Illustrate the realization of a phase comparator by using amplitude comparator.	5	L3	2	1,2

- c) The current rating of an IDMT overcurrent relay is 5 A. Its plug setting is 150% and TMS is 0.8. This relay is connected to the system through a 400/5 A CT. The fault current on the primary side of the CT is 2400 A. Determine the operating time of the relay. For TMS=1, the characteristics of the relay are as follows:

PSM	2	4	5	8	10	20
Operating time (s)	10	5	4	3	2.8	2.4

3. a) Write a comparative note on isolator, load-break switch and a circuit breaker. 5 L3 2 1,2
 b) Describe the cut-off characteristics of a fuse with a neat plot. 6 L2 1 1,2
 c) Illustrate the working of a Buchholz relay with a neat figure. How does it succeed in detecting a fault in the incipient stage? 3 L2 1 1,2
 7 L2 2 1,2

Unit – II

4. a) Illustrate aliasing error and hence define Nyquist's sampling theorem. 8 L3 3 2
 b) Discuss percentage biased differential protection. Obtain its characteristics. Illustrate how this succeeds in overcoming the drawbacks of a simple differential relay. 8 L3 4 1,2
 5. a) Describe analog multiplexing employed in a digital relay with a schematic diagram. Why is this required? 8 L4 3 1,2
 b) Describe reactance relay and hence obtain its torque equation. List its advantage and disadvantage. 8 L3 4 1,2
 6. a) Draw the block diagram of a numerical relay and discuss the constituent blocks in brief. 8 L2 3 1,2
 b) A generator is protected by restricted-earth-fault relay. The generator ratings are 13.2 kV, 5 MVA. The percentage of winding protected against phase-to-ground fault is 88. The relay setting is such that it trips for 15% out-of-balance current. Calculate the resistance to be added in neutral-to-ground connection. 4 L3 4 1,2
 c) Illustrate the differential protection of a bus. 4 L2 4 1,2

Unit – III

7. a) What are surge arresters? Describe the construction and working of a Silicon Carbide arrester with a neat figure. 5 L2 5 1,2
 b) With neat circuit diagrams illustrate the following methods of system earthing: (i) solid earthing (ii) resistance earthing (iii) reactance earthing. Hence write a comparative note on these. 7 L2 5 1,2
 c) Discuss the recovery rate theory of arc extinction. 4 L2 5 1,2
 8. a) Derive an expression for restriking voltage and rate of rise of restriking voltage of a circuit breaker. 10 L2 5 1,2
 b) In a 400 kV system, the reactance and capacitance up to the location of circuit breaker are 8Ω and $0.025 \mu\text{F}$ respectively. A resistance of 600Ω is connected across the contacts of circuit breaker. Determine:
 (a) natural frequency of oscillation (b) damped frequency of oscillation
 (c) critical resistance. 6 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
